

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SALES ORBIS AI-BASED SALES PIPELINE INTELLIGENCE AND DEAL WIN PREDICTION PLATFORM SDG GOAL NO : 8 – DECENT WORK AND ECONOMIC GROWTH

SUPERVISOR

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INTRODUCTION

- Domain: Artificial Intelligence, Machine Learning, and B2B Procurement Intelligence.
- B2B procurement involves managing suppliers and opportunities across multiple stages.
- Traditional procurement systems mainly provide descriptive insights with limited predictive capabilities.
- The proposed platform uses AI/ML to predict procurement outcomes and identify potential risks.
- The solution provides data-driven supplier intelligence and improved decision support.

PROBLEM STATEMENT

- B2B procurement involves managing numerous suppliers and opportunities, making manual evaluation and tracking inefficient.
- Traditional procurement systems store data but lack predictive insights for supplier selection and outcome prediction.
- Supplier evaluation and opportunity prioritization often depend on subjective judgment rather than data-driven analysis.
- Supplier risks and project-related issues are often identified late, reducing the scope for timely intervention.
- There is a need for an AI-driven platform that predicts procurement outcomes, identifies risks early, and supports better supplier decision-making.

PROJECT OVERVIEW & OBJECTIVES/SCOPE

- AI-based platform that predicts supplier win probability and forecasts purchasing value from historical purchasing data.
- Approach: **Data Preparation → Feature Engineering → Predictive Modeling → Forecasting → Report**

Objectives

- Predict the probability that a supplier will win a contract
- Identify the key factors driving purchasing outcomes
- Forecast expected purchasing value with confidence ranges
- Present predictions and risk levels through a detailed report

Scope

- Phase 1 (Current): Dataset validation, feature engineering, purchasing value forecasting, report generation
- Phase 2 (Planned): Next-best-action recommendations, real-time data feeds

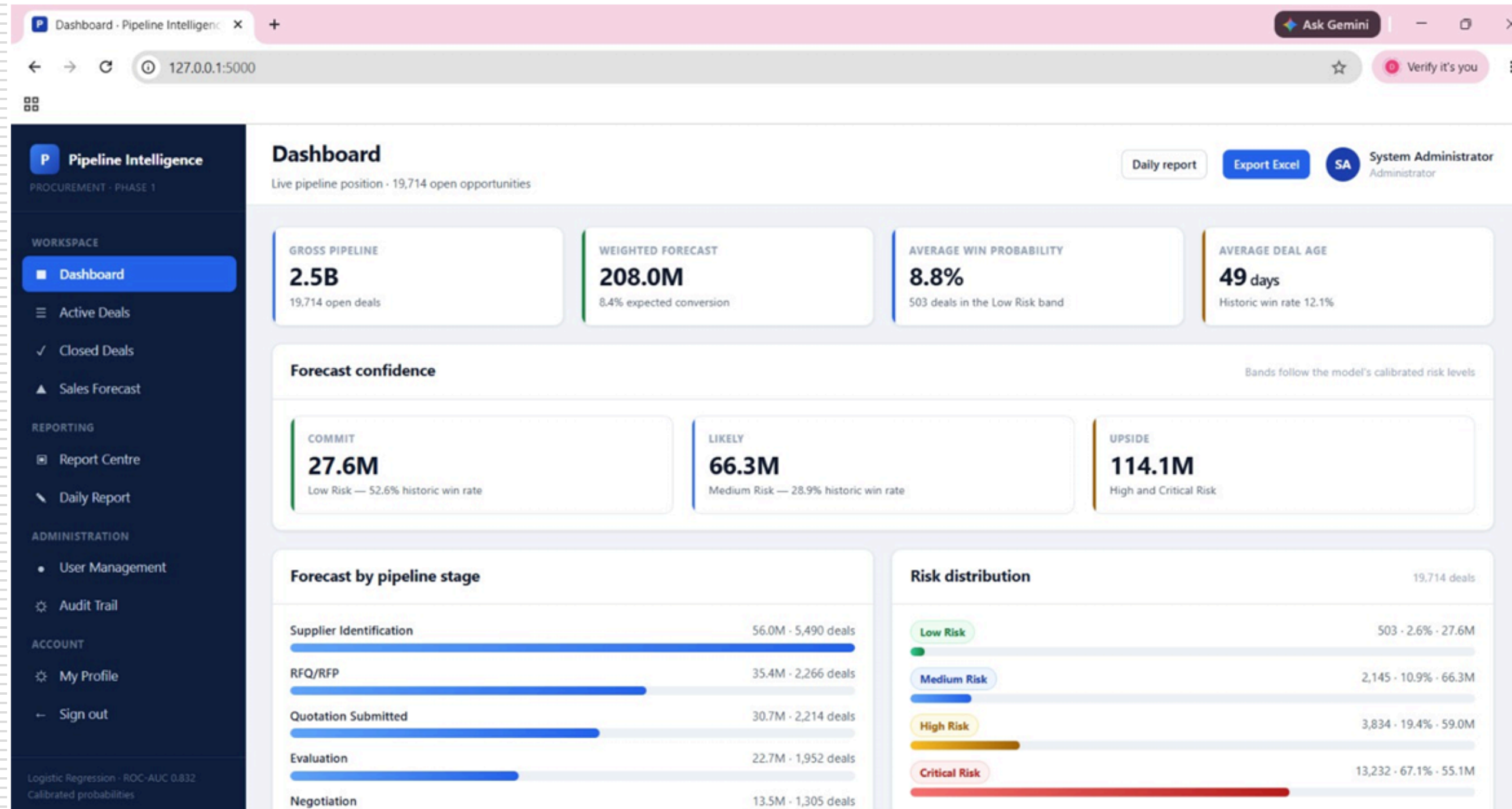
SYSTEM ARCHITECTURE MODULES

The proposed SalesOrbis Procurement Intelligence System is organized into six key modules that support procurement monitoring, prediction, analysis, forecasting, and administrative operations:

- Procurement Intelligence Dashboard
- Pipeline Analytics & Opportunity Prioritization
- Active Deal Intelligence & Prediction
- Forecasting & Confidence Analysis
- Closed Deal & Risk-Band Analysis
- Reporting, User Management & Audit

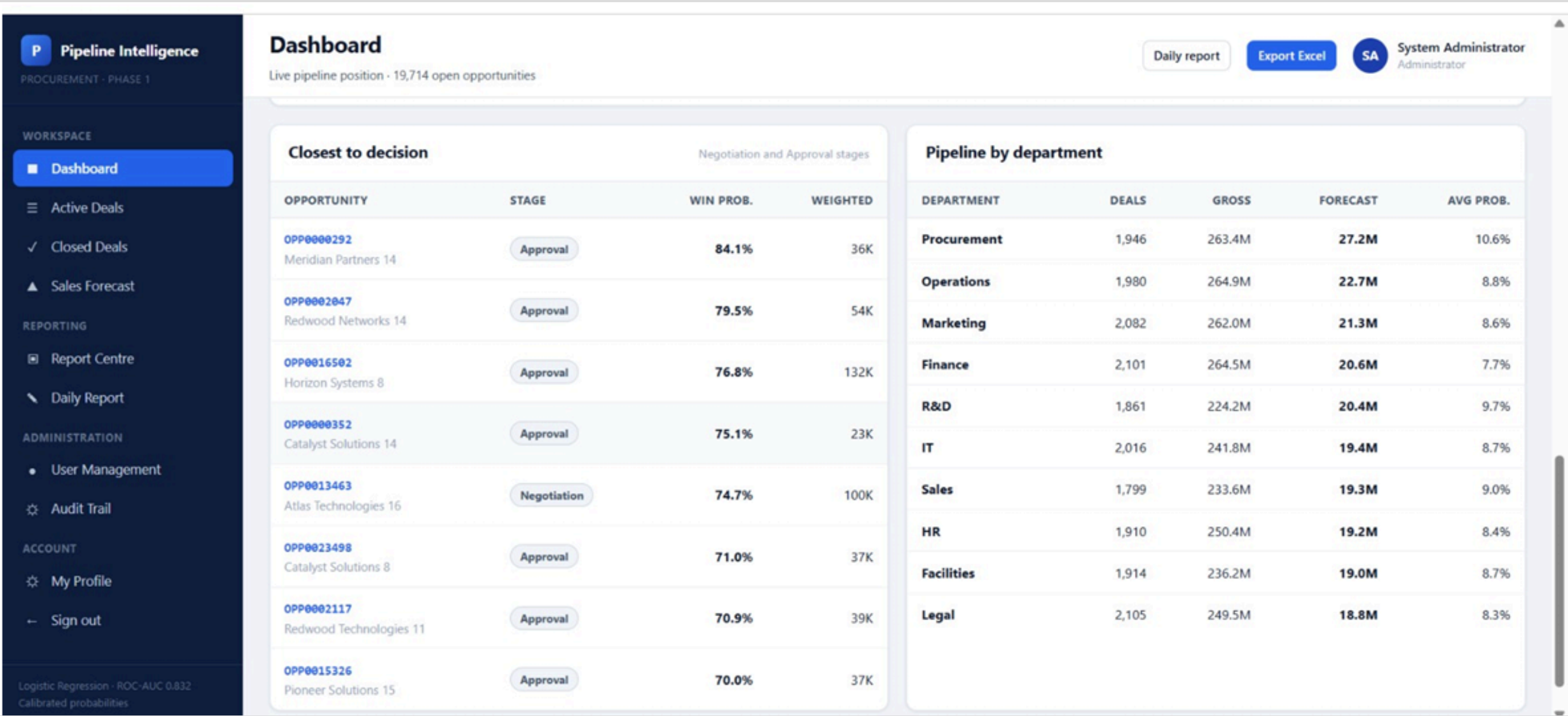
1. PROCUREMENT INTELLIGENCE DASHBOARD

- Provides an overall view of the procurement pipeline, including pipeline value, win probability, risk, and priority opportunities.



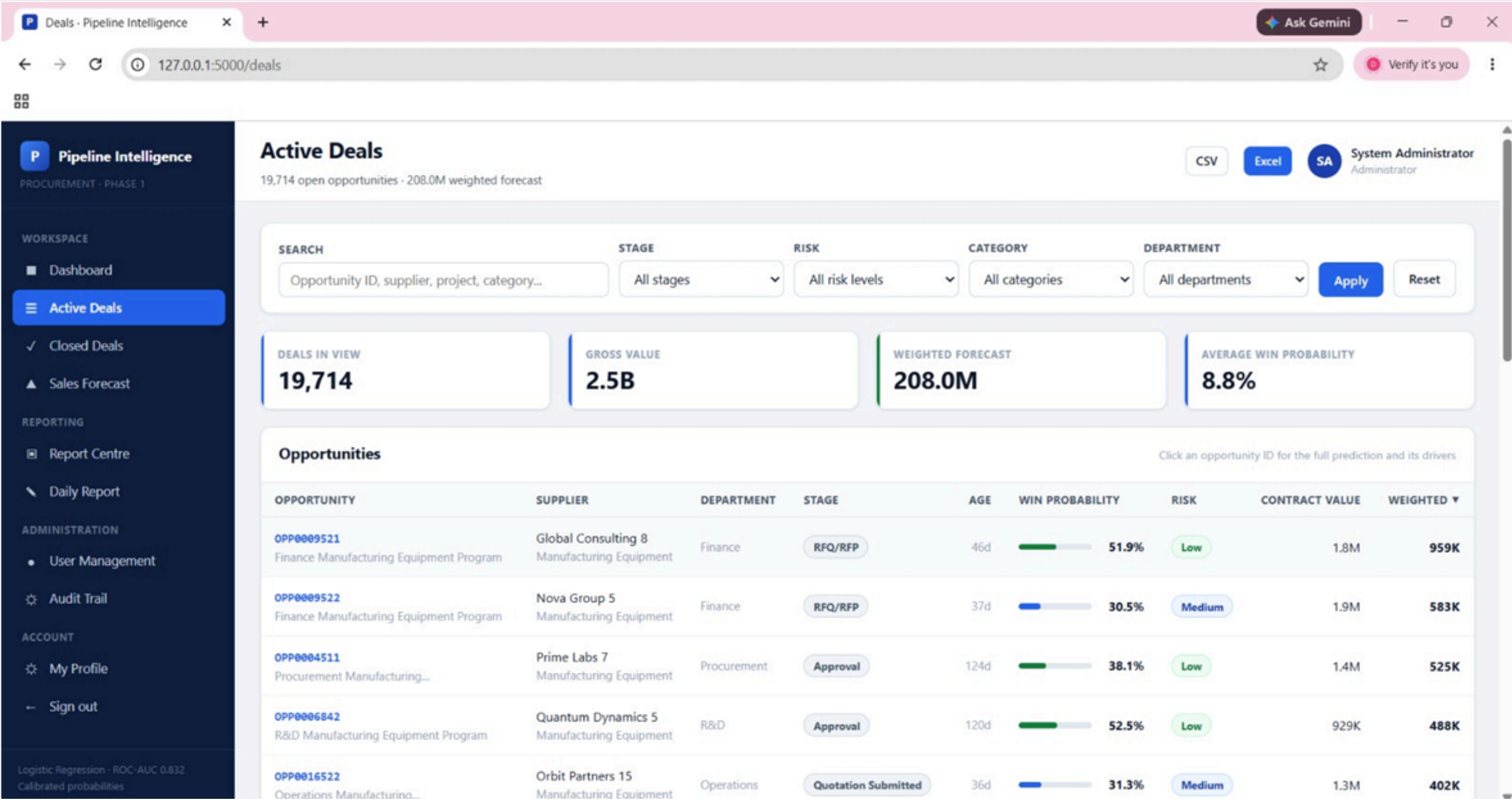
2. PIPELINE ANALYTICS & OPPORTUNITY PRIORITIZATION

- Analyzes procurement opportunities and helps prioritize them based on predicted outcomes, risk, and business value.



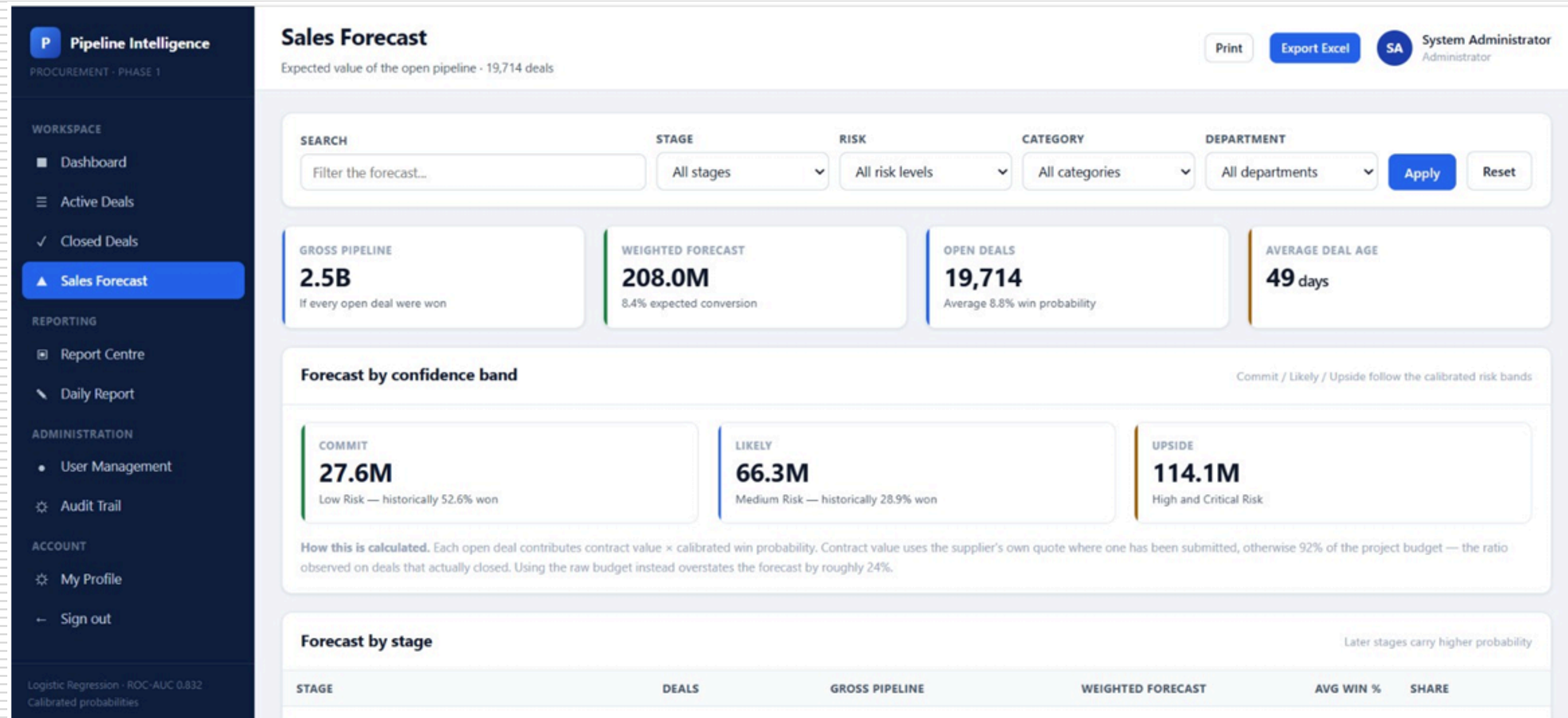
3. ACTIVE DEAL INTELLIGENCE & PREDICTION

- Displays active procurement opportunities with supplier details, procurement stage, predicted win probability, risk level, and contract value.



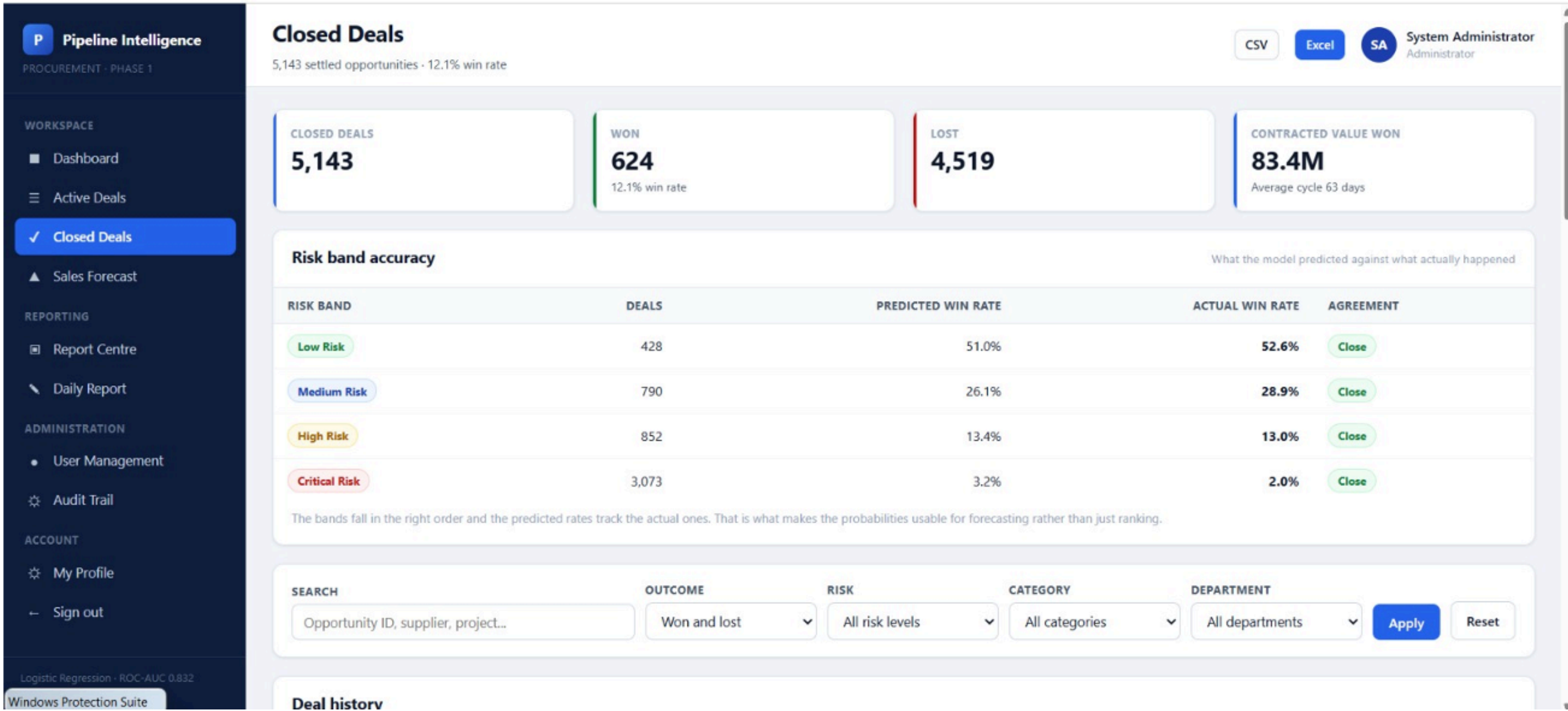
4. FORECASTING & CONFIDENCE ANALYSIS

- Generates procurement forecasts across stages, departments, categories, regions, and risk bands while showing forecast confidence.



5.CLOSED DEAL & RISK-BAND ANALYSIS

- Reviews historical procurement outcomes by comparing predicted win patterns, actual results, contract values, and risk bands.



EXISTING SYSTEM

- Traditional CRM systems are used for sales tracking.
- Forecasting is based on manual judgment and experience.
- Opportunity prioritization is not data-driven.
- At-risk deals are identified reactively.
- Supplier risks and performance issues are often identified reactively.
- Procurement decisions lack predictive, data-driven insights.

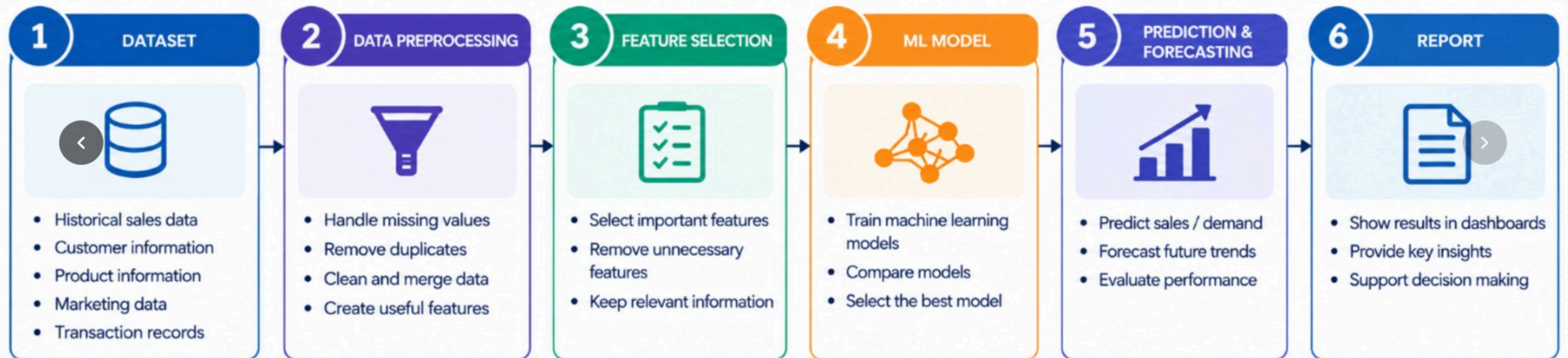
METHODOLOGY

- **Approach:** Dynamic supplier award prediction using procurement and supplier information available at each snapshot date.
- **Data Flow:** Dataset Validation → Data Preprocessing → Feature Engineering → Leakage Control → Model Training → Probability Calibration → Risk Assessment.
- **Tools & Technologies**
- **Data Processing:** The leakage-safe Gold ML dataset is used for supervised training and evaluation.
- **Feature Engineering:** Project, supplier, commercial, evaluation, pipeline, engagement, and competitive features are analysed.
- **Models:** Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost.
- **Evaluation:** ROC-AUC, PR-AUC, F1 Score, Brier Score, and calibration performance.
- **Final Output:** Award probability, prediction, risk level, and expected procurement value.

SYSTEM ARCHITECTURE

AI-Powered Sales Forecasting Pipeline

From Data to Actionable Insights



END-TO-END USE CASE – LIVE DEMO

Use Case Step	Module(s) Involved	Status (Live / Mocked / Not Built)	Notes
1. B2B Procurement Dataset Preparation	Data Generation & Preprocessing	Live	Procurement and supplier intelligence dataset prepared
2. Procurement Forecasting	Forecasting Module	Live	Forecasts expected procurement value and outcomes
3. Supplier Award Prediction	Machine Learning Prediction Model	Live	Predicts the probability of supplier award
4. User Engineering	User Interface & User Management	Live	Provides an interactive platform for users to access insights

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
Chahal et al., 2025	The Use of Machine Learning in Business Process Management	Systematic analysis of ML integration into core business operations, CRM, and workflow optimization	Demonstrates the cross-functional value and operational efficiency gains of ML-driven automation in enterprise processes
Doody, 2020	Machine Learning Techniques for Sales Forecasting in Business	Comparative evaluation of ML regressors and time-series models for revenue and demand projection	Validates that machine learning models outperform standard statistical methods in capturing complex market trends
Casas et al., 2020	Predictive Models for Customer Relationship Management (CRM)	Supervised classification and predictive analytics applied to lifecycle and churn management	Confirms predictive modeling directly improves sales cycle and pipeline prioritization
Kalia, 2020	Sales Lead Prioritization Using Machine Learning Approaches	Evaluation of classification algorithms (Random Forest, XGBoost, Logistic Regression) on B2B lead datasets	Proves algorithmic lead scoring optimizes conversion rates and streamlines pipeline prioritization
Sarcouncil Research, 2025	Smart Opportunity Scoring in Salesforce Sales Cloud: A Machine Learning Approach	Implementation of predictive opportunity scoring models integrated directly into Salesforce Sales Cloud	Shows that embedding ML scoring directly into CRM workflows increases deal conversion and sales rep productivity

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Bose et al., 2023	Advanced Neural and Ensemble Approaches for Enterprise Sales Forecasting	Ensemble learning combining gradient boosting with deep neural networks for multi-period sales forecasting	Proves hybrid/ensemble ML models provide superior robustness against volatility compared to standard regression
Schmid et al., 2000	Data Mining and Machine Learning in CRM Architectures	Classical architectural overview and algorithmic benchmark for data mining in customer databases	Establishes the structural foundation for integrating data mining pipelines into large-scale relational CRMs
Tang et al., 2025	Quantitative Forecasting in Financial and Customer Operations	Advanced quantitative econometrics and ML-based time-series forecasting across financial operations	Shows quantitative ML algorithms generalize robustly to multivariate, high-noise financial and sales datasets
Pang et al., 2023	Deep Learning Applications in Sales Opportunity and Lead Scoring	End-to-end deep learning framework utilizing tabular feature embeddings to predict deal closure rates	Demonstrates deep tabular networks capture nonlinear feature interactions better than baseline classifiers
Bohanec et al., 2019	Combining Rule-Based Systems and Machine Learning for B2B Sales Forecasting	Hybrid expert system combining qualitative rule-based reasoning with quantitative ML forecasting	Validates that combining domain knowledge heuristics with ML improves sales pipeline win/loss predictions

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Pazgan & Czerwonka, 2025	The Use of Machine Learning in Business Process Management	Systematic analysis of ML integration into core business operations, CRM, and workflow optimization	Demonstrates the cross-functional value and operational efficiency gains of ML-driven automation in enterprise processes
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Publication Status - Applied

Submission Summary

Conference Name

International Conference on Computation, Communication, AND Intelligent Systems in Engineering

Track Name

Track 01: Computational Intelligence & Advanced Data Systems

Paper ID

519

Paper Title

Sales Orbis AI-Based Sales Pipeline Intelligence and Deal Win Prediction Platform

Abstract

The current study is concerned with the problem of estimating the probabilities of awarding contracts to suppliers in competitive procurement, where there exist several suppliers who

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Primary Subject Area

AI, Machine Learning & Deep Learning

Submission Files

<https://cmt3.research.microsoft.com/ICONCISE2027/Submission/Summary/519>

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Conference Management Toolkit - Submission Summary

Final Paper 1.pdf (629.1 Kb, 2026-10-04 21:27:03 GMT+5:30)